The Musk Effect: Changes in the Twitter's Information Quality and Partisan Composition

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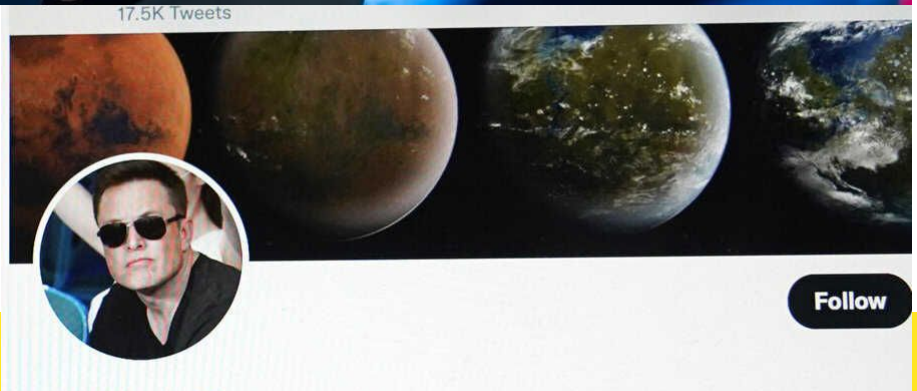
Elon Musk
@elonmusk
Let's make Twitter maximum fun!
2:53 AM · Apr 28, 2022 · Twitter for iPhone

ye @kanyewest · Jul 4, 2020
We must now realize the promise of America by trusting God, unifying our vision and building our future. I am running for president of the United States ! #2020VISION
99.3K
460.6K
1M
Elon Musk
@elonmusk
Replying to @kanyewest
You have my full support!
8:45 PM · Jul 4, 2020 · Twitter for iPhor
Elon Musk
@elonmusk
Short shorts coming soon to Tesla merch
11:30 PM · Aug 10, 2018 · Twitter
Elon Musk
@elonmusk
The rumor that Bill Gates & I are lovers is completely untrue
2:55 AM · Jul 30, 2020 · Twitter for iPhone
Elon Musk
@elonmusk
Joined June 2009


17.5K Tweets
Follow

ALL FOR $8
SCREAMING
Elon Musk
@elonmusk
Living the dream

Motivation

CBS NEWS
MONEYWATCH
Musk fires outs
moderators wh
Twitter

Elon Musk
@elonmusk · Follow
The people have spoken.
Trump will be reinstated.
Vox Populi, Vox
Elon Musk
Reinstate former
7:53 PM · Nov 19, 2
912.9K

Elon Musk
@elonmusk
To independent-minded voters:
Shared power curbs the worst excesses of both parties, therefore I recommend voting for a Republican Congress, given that the Presidency is Democratic.
10:22 AM · Nov 7, 2022
53.9K Reposts 28.5K Quotes 392.2K Likes 2,420 Bookmarks

Motivation

The New York Times

Hate Speech's Rise on Twitter Is Unprecedented, Researchers Find

Problematic content and formerly barred accounts have increased sharply in the short time since Elon Musk took over, researchers said.

NewsGuard

Twitter Misinformation Superspreaders See Huge Spike in Engagement Post-Acquisition by Elon Musk

Twitter's most popular NewsGuard Red-Rated untrustworthy accounts garnered 57.04% increase in engagement in the week following the change in ownership

(Un)Natural Experiment

1. The effect of the ownership of the platform and its governance

Meaningful measures of human society in the twenty-first century

https://doi.org/10.1038/s41586-021-03660-7
Received: 3 March 2021

David Lazer[1,2], Eszter Hargittai[3], Deen Freelon[4], Sandra Gonzalez-Bailon[5], Kevin Munger[6], Katherine Ognyanova[7] & Jason Radford[1]

2. What happens to our Town Square

Who Leads? Who Follows? Measuring Issue Attention and Agenda Setting by Legislators and the Mass Public Using Social Media Data

PABLO BARBERÁ *University of Southern California*
ANDREU CASAS *New York University*
JONATHAN NAGLER *New York University*
PATRICK J. EGAN *New York University*
RICHARD BONNEAU *New York University*
JOHN T. JOST *New York University*
JOSHUA A. TUCKER *New York University*

Motivation

Elon Musk
@elonmusk

In the past I voted Democrat, because they were (mostly) the kindness party.

But they have become the
support them and will vot

Now, watch their dirty tric

2:44 PM · May 18, 2022

120.2K Reposts **32.8K** Quot

Elon Musk
@elonmusk

To independent-minded voters:

Shared power curbs the worst excesses of both parties, therefore I recommend voting for a Republican Congress, given that the Presidency is Democratic.

10:22 AM · Nov 7, 2022

53.9K Reposts **28.5K** Quotes **392.2K** Likes **2,420** Bookmarks

Research Questions

1. How has the information quality on Twitter changed?
2. How has the demographic composition of Twitter changed?

Data and Methods

1. Decahose
2. Twitter Panel
3. Covid States Project

Decahose

Random 10% of Twitter:

- Advantages:
  - Overall Information Quality
  - Measuring misinformation

- Disadvantages:
  - Real persons vs social bots
  - US public opinion

Twitter Panel

1.6M Twitter users matched to voter file unique names within a city, state:

- Advantages:
  - Real person with known demographics (partisanship) not bots or organizations
  - Representative of User voter on Twitter Grinberg et al. 2019,Hughes et al., 2021, Shugars et al., 2021

- Disadvantages:
  - Fixed, no new entrants

Covid States Project>CHIP50

A 21 waves of large-scale online non-probability survey:

- Bimonthly btw April 2020 and July 2023.
- Every US State and DC
- 20,000 respondents per wave.

- Advantages:
  - Dynamic sample: study self-reported Twitter users in our sample

- Disadvantages:
  - Self Report: memory bias, social desirability bias

Measuring Information Quality

Domain	Fact-checker	AFM	Iffy	MBFC	Lasser	NewsGuard	PC1
reuters	0.67	0.96	1.00	1.00	1.00	-	1.00
afp	0.67	0.95	1.00	0.92	1.00	-	0.95
economist	0.66	0.87	1.00	0.92	1.00	-	0.93
usatoday	0.73	0.89	1.00	0.92	1.00	-	0.90
chicagotribune	0.58	0.90	1.00	0.83	1.00	0.92	0.87
nytimes	1.00	0.81	1.00	0.83	1.00	1.00	0.86
washingtonpost	1.00	0.81	1.00	0.75	1.00	-	0.82
theguardian	0.77	0.78	1.00	0.67	1.00	-	0.75
huffingtonpost	0.52	0.73	1.00	0.83	1.00	-	0.72
commondreams	0.03	0.60	1.00	0.75	1.00	-	0.54
foxnews	0.48	0.61	0.00	0.58	1.00	0.57	0.53
rawstory	0.10	0.53	1.00	0.58	1.00	-	0.46
breitbart	0.18	0.51	0.00	0.50	0.00	0.50	0.30
newsmax	0.14	0.49	0.00	0.50	0.00	-	0.29
zenith.news	0.05	0.47	0.00	0.42	0.00	-	0.23
healthnutnews	0.08	0.48	0.00	0.06	0.00	-	0.15
whatdoesitmean	0.00	0.37	1.00	0.11	0.00	-	0.15
beforeitsnews	0.00	0.01	0.00	0.42	0.00	0.00	0.06
infowars	0.03	0.16	0.00	0.11	0.00	-	0.05
naturalnews	0.07	0.03	0.00	0.11	0.00	-	0.00

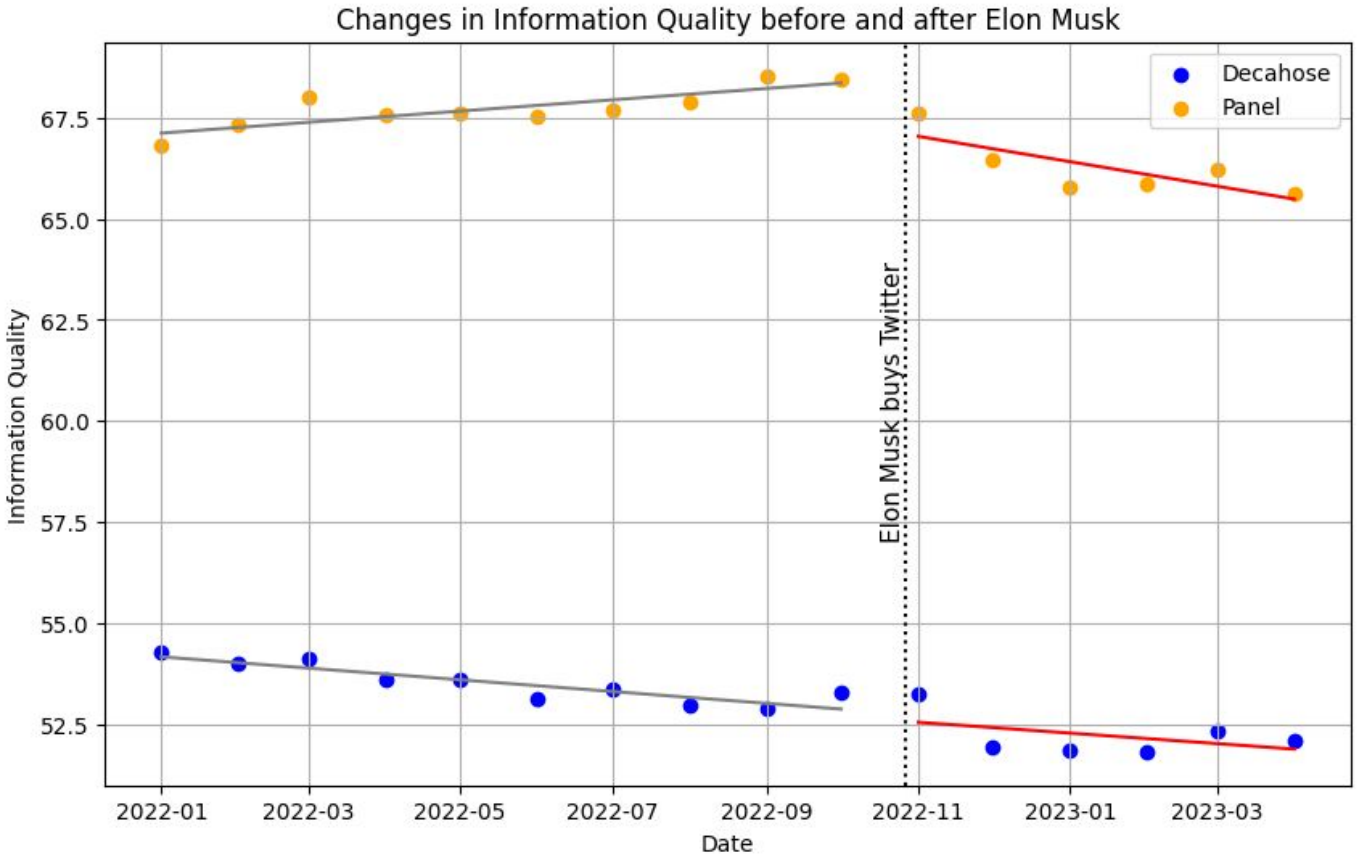
Note: AFM: Ad Fontes Media. Iffy: Iffy index of unreliable sources. Several NewsGuard ratings have been redacted (-) because NewsGuard only permits the publication of five example ratings. MBFC: Media Bias/Fact Check. Lasser: Lasser: [17]. PC1: first principal component.

High level of correspondence across different news domain quality rating sets

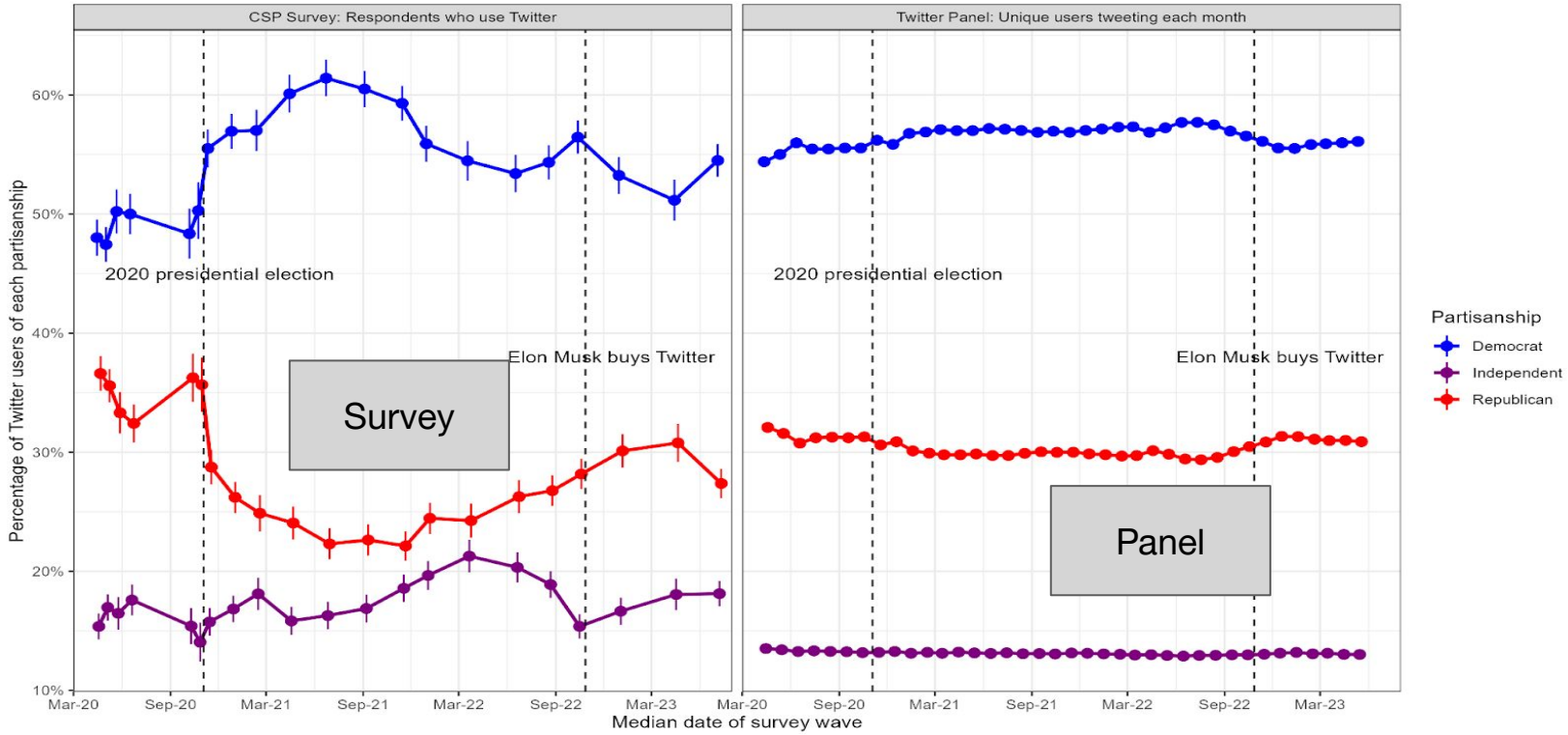
PNAS Nexus

Hause Lin[1,2,3*], Jana Lasser[4,5], Stephan Lewandowsky[6,7], Rocky Cole[8^], Andrew Gully[8^], David G. Rand[2,9], Gordon Pennycook[1,3,10]

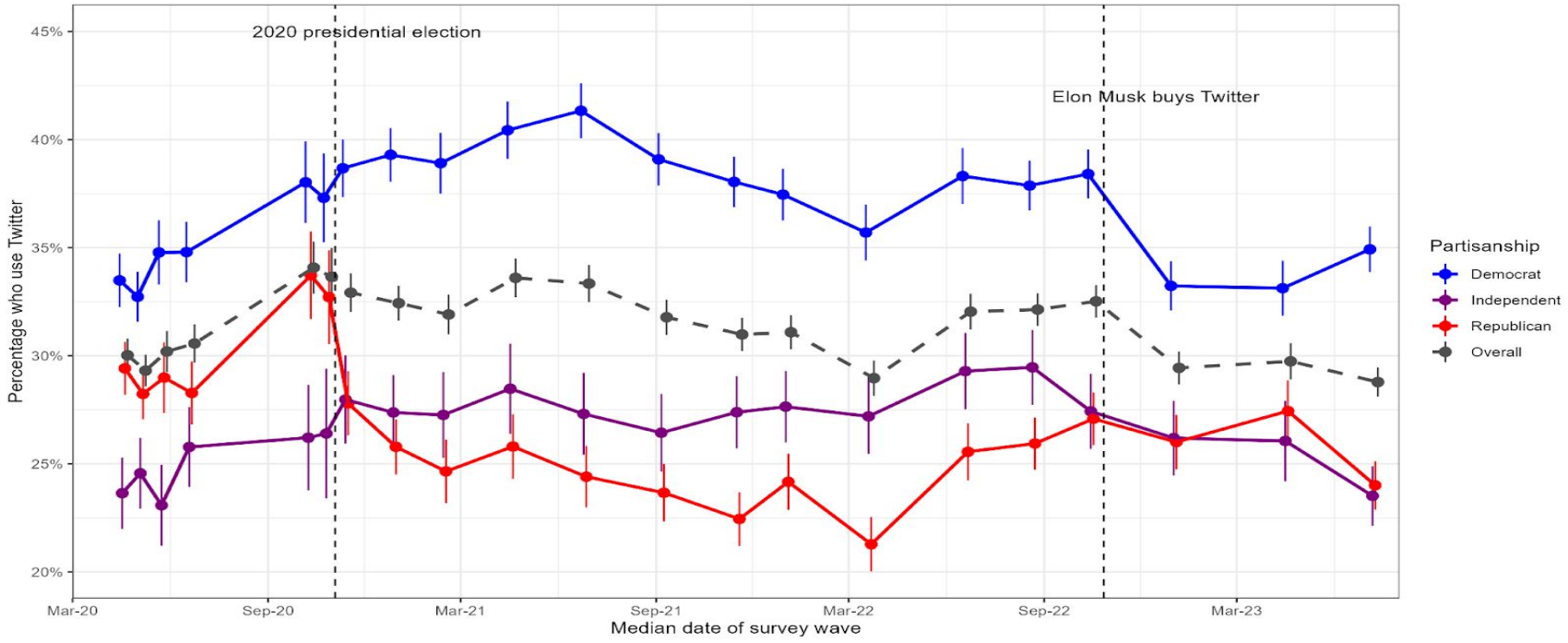
RQ1: Information Quality



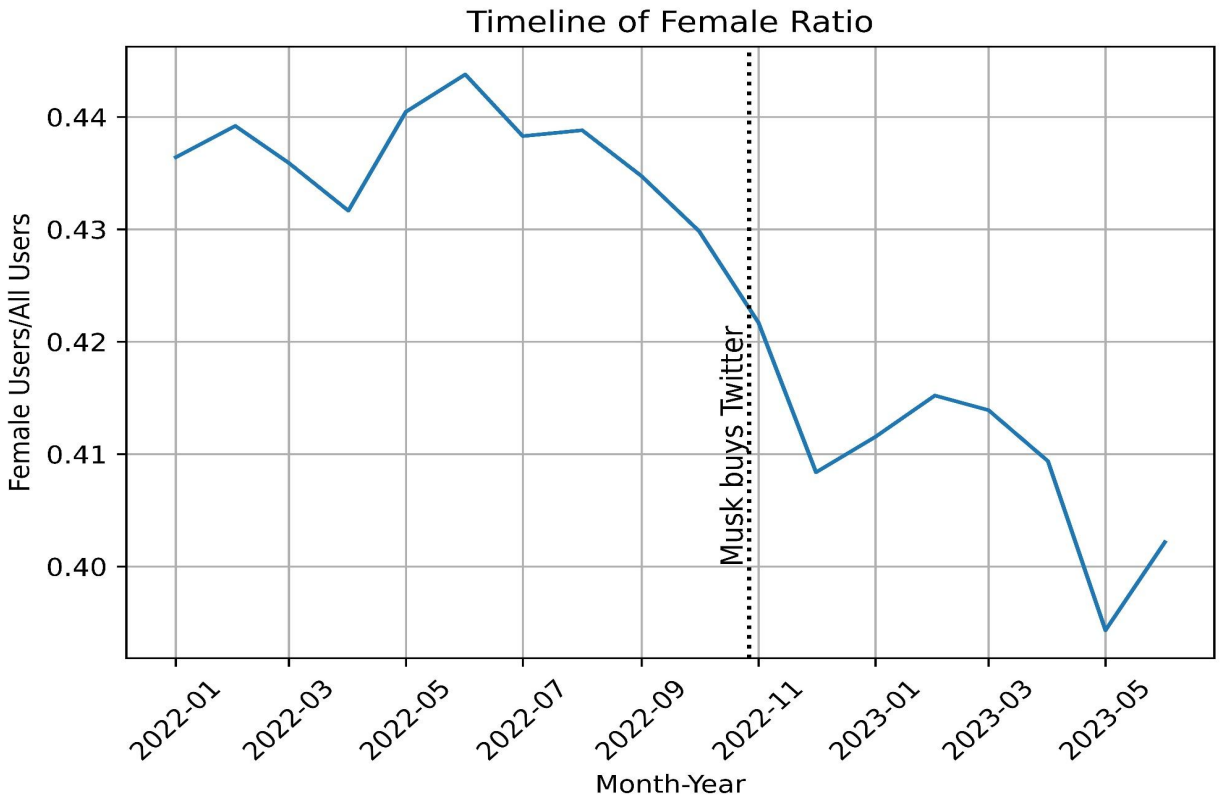
RQ2: User Demographics: Partisanship



Democrats left the platform…



RQ2: User Demographics: Gender



Takeaways

Governance of the platform and platform ownership matters…

Decrease in information quality

Change in user demographics

Questions and Feedbacks

Thank you

Looking for Internships :)

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@burakozturan

Data (i): Twitter Users

10%

>406 million
(Decahose
1/2014-3/2017)

>322 million
(active)

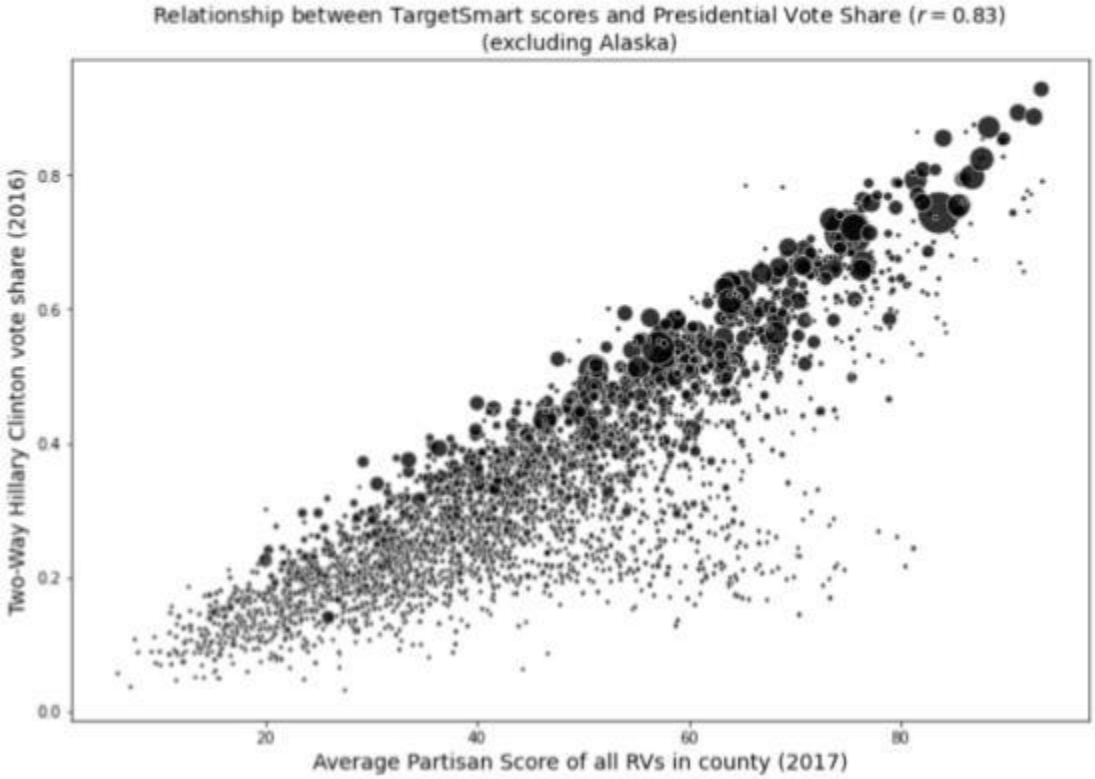
Stefan McCabe
@mccabe_s

Econlib
@Econlib

290 million
(have names)

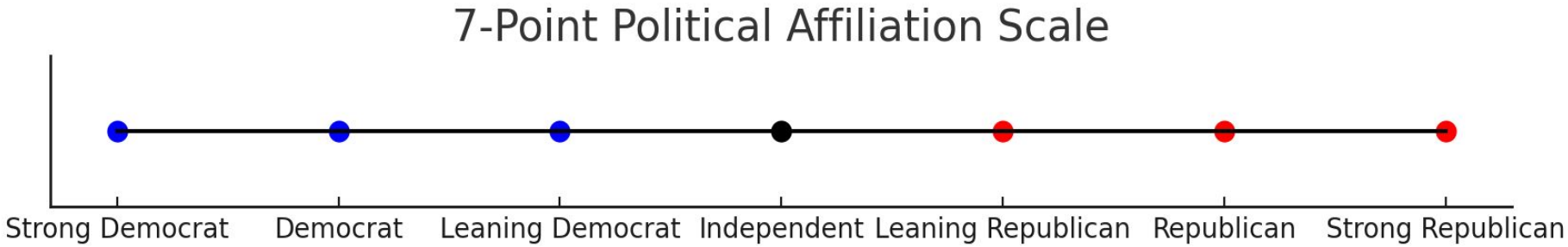
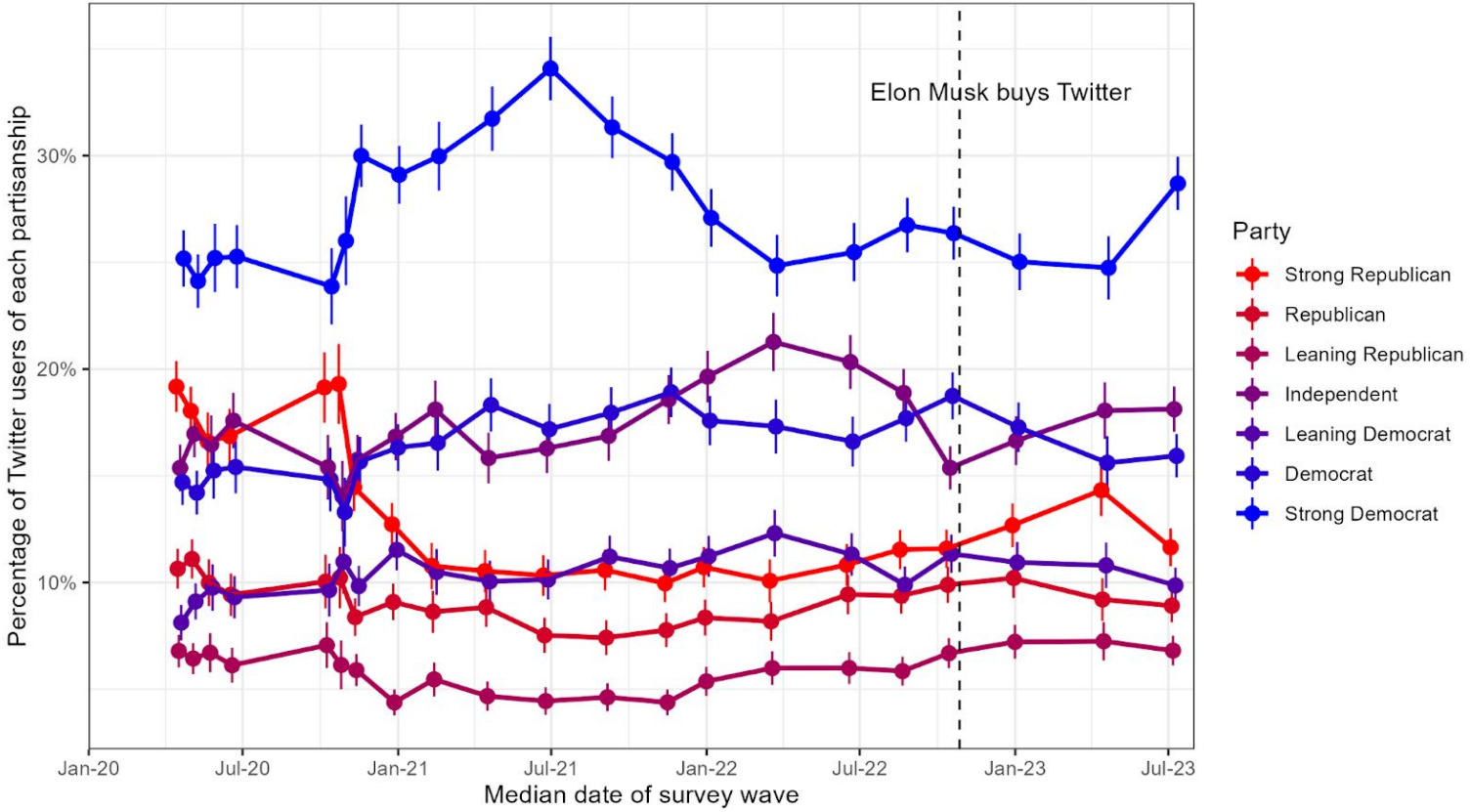
1.5 million
(linked to
voter data)

TargetSmart Partisanship score



Covid States Project

1. We asked respondents if they ever use Twitter
2. We also asked: "Do you identify as a Republican, Democrat, Independent, or another party?"
   a. If Independent/Other: Do you lean more towards Republican, Democrat, or neither?"
   b. For Republicans/Democrats: Do you consider yourself strong or not very strong in your affiliation?"



Strong Partisanship